

Phonetics

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*For the study of phonemes, or how languages organize sounds, see **Phonology**. For the method of teaching reading and writing, see **Phonics**. For other uses, see **Phonetics (disambiguation)**.*

Phonetics is a branch of **linguistics** that studies how humans produce and perceive sounds or, in the case of **sign languages**, the equivalent aspects of sign.^[1] Linguists who specialize in studying the physical properties of speech are **phoneticians**. The field of phonetics is traditionally divided into three sub-disciplines: **articulatory phonetics**, **acoustic phonetics**, and **auditory phonetics**. Traditionally, the minimal linguistic unit of phonetics is the **phone**—a speech sound in a language which differs from the phonological unit of **phoneme**; the phoneme is an abstract categorization of phones and it is also defined as the smallest unit that discerns meaning between sounds in any given language.^[2]

Phonetics deals with two aspects of human speech: production (the ways humans make sounds) and perception (the way speech is understood). The **communicative modality** of a language describes the method by which a language produces and perceives languages. Languages with oral-aural modalities such as English produce speech orally and perceive speech aurally (using the ears). Sign languages, such as **Australian Sign Language** (Auslan) and **American Sign Language** (ASL), have a manual-visual modality, producing speech manually (using the hands) and perceiving speech visually. ASL and some other sign languages have in addition a manual-manual dialect for use in **tactile signing** by **deafblind** speakers where signs are produced with the hands and perceived with the hands as well.

History [edit]

Antiquity [edit]

The first known study of phonetics was undertaken by **Sanskrit** grammarians as early as the 6th century BCE.^[3] The Hindu scholar **Pāṇini** is among the most well known of these early investigators. His four-part grammar, written c. 350 BCE, is influential in modern linguistics and still represents "the most complete generative grammar of any language yet written".^[4] His grammar formed the basis of modern linguistics and described several important phonetic principles, including voicing. This early account described resonance as being produced either by tone, when vocal folds are closed, or noise, when vocal folds are open. The phonetic principles in the grammar are considered "primitives" in that they are the basis for his theoretical analysis rather than the objects of theoretical analysis themselves, and the principles can be inferred from his system of phonology.^[5]

The Sanskrit study of phonetics is called ***Shiksha***, which the 1st-millennium BCE **Taittiriya Upanishad** defines as follows:

Om! We will explain the Shiksha.
Sounds and accentuation, Quantity (of vowels) and the expression (of consonants),
Balancing (Saman) and connection (of sounds), So much about the study of Shiksha.
|| 1 |

Taittiriya Upanishad 1.2, Shikshavalli, translated by Paul Deussen^[6].

Modern [edit]

Advancements in phonetics after Pāṇini and his contemporaries were limited until the modern era, save some limited investigations by Greek and Roman grammarians. In the millennia between Indic grammarians and modern phonetics, the focus shifted from the difference between spoken and written language, which was the driving force behind Pāṇini's account, and began to focus on the physical properties of speech alone. Sustained interest in phonetics began again around 1800 CE with the term "phonetics" being first used in the present sense in 1841.^{[7][3]} With new developments in medicine and the development of audio and visual recording devices, phonetic insights were able to use and review new and more detailed data. This early period of modern phonetics included the development of an influential phonetic alphabet based on articulatory positions by **Alexander Melville Bell**. Known as **visible speech**, it gained prominence as a tool in the **oral education of deaf children**.^[3]

Before the widespread availability of audio recording equipment, phoneticians relied heavily on a tradition of practical phonetics to ensure that transcriptions and findings were able to be consistent across phoneticians. This training involved both ear training—the recognition of speech sounds—as well as production training—the ability to produce sounds. Phoneticians were expected to learn to recognize by ear the various sounds on the **International Phonetic Alphabet** and the IPA still tests and certifies speakers on their ability to accurately produce the phonetic patterns of English (though they have discontinued this practice for other languages).^[8] As a revision of his visible speech method, Melville Bell developed a description of vowels by height and backness resulting in 9 **cardinal vowels**.^[9] As part of their training in practical phonetics, phoneticians were expected to learn to produce these cardinal vowels to anchor their perception and transcription of these phones during fieldwork.^[8] This approach was critiqued by **Peter Ladefoged** in the 1960s based on experimental evidence where he found that cardinal vowels were auditory rather than articulatory targets, challenging the claim that they represented articulatory anchors by which phoneticians could judge other articulations.^[10]

Production [edit]

*Main article: **Language production***

Language production consists of several interdependent processes which transform a nonlinguistic message into a spoken or signed linguistic signal. Linguists debate whether the process of language production occurs in a series of stages (serial processing) or whether production processes occur in parallel. After identifying a message to be linguistically encoded, a speaker must select the individual words—known as **lexical items**—to represent that message in a process called lexical selection. The words are selected based on their meaning, which in linguistics is called **semantic** information. Lexical selection activates the word's **lemma**, which contains both semantic and grammatical information about the word.^{[11][a]}

After an utterance has been planned,^[b] it then goes through phonological encoding. In this stage of language production, the mental representation of the words are assigned their phonological content as a sequence of **phonemes** to be produced. The phonemes are specified for articulatory features which denote particular goals such as closed lips or the tongue in a particular location. These phonemes are then coordinated into a sequence of muscle commands that can be sent to the muscles, and when these commands are executed properly the intended sounds are produced.^[13] Thus the process of production from message to sound can be summarized as the following sequence:^[c]

- Message planning
- Lemma selection
- Retrieval and assignment of phonological word forms
- Articulatory specification
- Muscle commands
- Articulation
- Speech sounds

Place of articulation [edit]

*Main article: **Place of articulation***

Sounds which are made by a full or partial constriction of the vocal tract are called **consonants**. Consonants are pronounced in the vocal tract, usually in the mouth, and the location of this constriction affects the resulting sound. Because of the close connection between the position of the tongue and the resulting sound, the place of articulation is an important concept in many subdisciplines of phonetics.

Sounds are partly categorized by the location of a constriction as well as the part of the body doing the constricting. For example, in English the words ***fought*** and ***thought*** are a **minimal pair** differing only in the organ making the construction rather than the location of the construction. The "f" in ***fought*** is a labiodental articulation made with the bottom lip against the teeth. The "th" in ***thought*** is a linguodental articulation made with the tongue against the teeth. Constrictions made by the lips are called **labials** while those made with the tongue are called lingual.

Constrictions made with the tongue can be made in several parts of the vocal tract, broadly classified into coronal, dorsal and radical places of articulation. **Coronal** articulations are made with the front of the tongue, **dorsal** articulations are made with the back of the tongue, and **radical** articulations are made in the **pharynx**.^[14] These divisions are not sufficient for distinguishing and describing all speech sounds.^[14] For example, in English the sounds [s] and [ʃ] are both coronal, but they are produced in different places of the mouth. To account for this, more detailed places of articulation are needed based upon the area of the mouth in which the constriction occurs.^[15]

Labial [edit]

Articulations involving the lips can be made in three different ways: with both lips (bilabial), with one lip and the teeth, so they have the lower lip as the active articulator and the upper teeth as the passive articulator^[16] (labiodental), and with the tongue and the upper lip (linguolabial).^[17] Depending on the definition used, some or all of these kinds of articulations may be categorized into the class of **labial articulations**. **Bilabial consonants** are made with both lips. In producing these sounds the lower lip moves farthest to meet the upper lip, which also moves down slightly,^[18] though in some cases the force from air moving through the aperture (opening between the lips) may cause the lips to separate faster than they can come together.^[19] Unlike most other articulations, both articulators are made from soft tissue, and so bilabial stops are more likely to be produced with incomplete closures than articulations involving hard surfaces like the teeth or palate. Bilabial stops are also unusual in that an articulator in the upper section of the vocal tract actively moves downward, as the upper lip shows some active downward movement.^[20] **Linguolabial consonants** are made with the blade of the tongue approaching or contacting the upper lip. Like in bilabial articulations, the upper lip moves slightly towards the more active articulator. Articulations in this group do not have their own symbols in the International Phonetic Alphabet, rather, they are formed by combining an apical symbol with a diacritic implicitly placing them in the coronal category.^{[21][22]} They exist in a number of languages indigenous to **Vanuatu** such as **Tangoa**.

Labiodental consonants are made by the lower lip rising to the upper teeth. Labiodental consonants are most often **fricatives** while labiodental nasals are also typologically common.^[23] There is debate as to whether true labiodental **plosives** occur in any natural language,^[24] though a number of languages are reported to have labiodental plosives including **Zulu**,^[25] **Tonga**,^[26] and **Shubi**.^[24]

Coronal [edit]

Part of a series on Phonetics
Subdisciplines
Acoustic · Articulatory · Auditory
Articulation
Places of articulation
Labial · Dental · Alveolar · Postalveolar · Palatal · Velar · Uvular · Laryngeal
Manners of articulation
Consonant · Plosive · Affricate · Fricative · Nasal · Approximant · Liquid · Lateral · Vowel
Airstream mechanisms
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Acoustics
Phonation (Voicing)
Fundamental frequency · Glottis · Pitch · Pitch accent · Source–filter model · Tone · Voicelessness
Phonation types
Modal · Creaky · Breathy
Perception
Acoustic cues · Aphasia · Categorical perception · Hearing · Neural encoding of sound · Prosody
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For some articulations. When articulations are voiced, the main source of noise is the periodic vibration of the vocal folds. Articulations like voiceless plosives have no acoustic source and are noticeable by their silence, but other voiceless sounds like fricatives create their own acoustic source regardless of phonation.

Phonation is controlled by the muscles of the larynx, and languages make use of more acoustic detail than binary voicing. During phonation, the vocal folds vibrate at a certain rate. This vibration results in a periodic acoustic waveform comprising a **fundamental frequency** and its harmonics. The fundamental frequency of the acoustic wave can be controlled by adjusting the muscles of the larynx, and listeners perceive this fundamental frequency as pitch. Languages use pitch manipulation to convey lexical information in tonal languages, and many languages use pitch to mark prosodic or pragmatic information.

For the vocal folds to vibrate, they must be in the proper position and there must be air flowing through the glottis.^[50] Phonation types are modeled on a continuum of glottal states from completely open (voiceless) to completely closed (glottal stop). The optimal position for vibration, and the phonation type most used in speech, modal voice, exists in the middle of these two extremes. If the glottis is slightly wider, breathy voice occurs, while bringing the vocal folds closer together results in creaky voice.^[62]

The normal phonation pattern used in typical speech is modal voice, where the vocal folds are held close together with moderate tension. The vocal folds vibrate as a single unit periodically and efficiently with a full glottal closure and no aspiration.^[63] If they are pulled farther apart, they do not vibrate and so produce voiceless phones. If they are held firmly together they produce a glottal stop.^[62]

If the vocal folds are held slightly further apart than in modal voicing, they produce phonation types like breathy voice (or murmur) and whispery voice. The tension across the vocal ligaments (**vocal cords**) is less than in modal voicing allowing for air to flow more freely. Both breathy voice and whispery voice exist on a continuum loosely characterized as going from the more periodic waveform of breathy voice to the more noisy waveform of whispery voice. Acoustically, both tend to dampen the first formant with whispery voice showing more extreme deviations.^[64]

Holding the vocal folds more tightly together results in a creaky voice. The tension across the vocal folds is less than in modal voice, but they are held tightly together resulting in only the ligaments of the vocal folds vibrating.^[e] The pulses are highly irregular, with low pitch and frequency amplitude.^[65]

Some languages do not maintain a voicing distinction for some consonants,^[f] but all languages use voicing to some degree. For example, no language is known to have a phonemic voicing contrast for vowels with all known vowels canonically voiced.^[g] Other positions of the glottis, such as breathy and creaky voice, are used in a number of languages, like **Jalapa Mazatec**, to contrast **phonemes** while in other languages, like English, they exist allophonically.

There are several ways to determine if a segment is voiced or not, the simplest being to feel the larynx during speech and note when vibrations are felt. More precise measurements can be obtained through acoustic analysis of a spectrogram or spectral slice. In a spectrographic analysis, voiced segments show a voicing bar, a region of high acoustic energy, in the low frequencies of voiced segments.^[66] In examining a spectral splice, the acoustic spectrum at a given point in time a model of the vowel pronounced reverses the filtering of the mouth producing the spectrum of the glottis. A computational model of the unfiltered glottal signal is then fitted to the inverse filtered acoustic signal to determine the characteristics of the glottis.^[67] Visual analysis is also available using specialized medical equipment such as ultrasound and endoscopy.^{[66][h]}

Vowels [edit]

Vowels are broadly categorized by the area of the mouth in which they are produced, but because they are produced without a constriction in the vocal tract their precise description relies on measuring acoustic correlates of tongue position. The location of the tongue during vowel production changes the frequencies at which the cavity resonates, and it is these resonances—known as **formants**—which are measured and used to characterize vowels.

Vowel height traditionally refers to the highest point of the tongue during articulation.^[68] The height parameter is divided into four primary levels: high (close), close-mid, open-mid, and low (open). Vowels whose height are in the middle are referred to as mid. Slightly opened close vowels and slightly closed open vowels are referred to as near-close and near-open respectively. The lowest vowels are not just articulated with a lowered tongue, but also by lowering the jaw.^[69]

While the IPA implies that there are seven levels of vowel height, it is unlikely that a given language can minimally contrast all seven levels. **Chomsky** and **Halle** suggest that there are only three levels,^[70] although four levels of vowel height seem to be needed to describe **Danish** and it is possible that some languages might even need five.^[71]

Vowel backness is dividing into three levels: front, central and back. Languages usually do not minimally contrast more than two levels of vowel backness. Some languages claimed to have a three-way backness distinction include **Nimboran** and **Norwegian**.^[72]

In most languages, the lips during vowel production can be classified as either rounded or unrounded (spread), although other types of lip positions, such as compression and protrusion, have been described. Lip position is correlated with height and backness: front and low vowels tend to be unrounded whereas back and high vowels are usually rounded.^[73] Paired vowels on the IPA chart have the spread vowel on the left and the rounded vowel on the right.^[74]

Together with the universal vowel features described above, some languages have additional features such as **nasality**, **length** and different types of phonation such as **voiceless** or **creaky**. Sometimes more specialized tongue gestures such as **rhoticity**, **advanced tongue root**, **pharyngealization**, **stridency** and frication are required to describe a certain vowel.^[75]

Manner of articulation [edit]

*Main article: **Manner of articulation***

Knowing the place of articulation is not enough to fully describe a consonant, the way in which the stricture happens is equally important. Manners of articulation describe how exactly the active articulator modifies, narrows or closes off the vocal tract.^[76]

Stops (also referred to as plosives) are consonants where the airstream is completely obstructed. Pressure builds up in the mouth during the stricture, which is then released as a small burst of sound when the articulators move apart. The velum is raised so that air cannot flow through the nasal cavity. If the velum is lowered and allows for air to flow through the nose, the result in a nasal stop. However, phoneticians almost always refer to nasal stops as just "nasals".^[76] **Affricates** are a sequence of stops followed by a fricative in the same place.^[77]

Fricatives are consonants where the airstream is made turbulent by partially, but not completely, obstructing part of the vocal tract.^[76] **Sibilants** are a special type of fricative where the turbulent airstream is directed towards the teeth,^[78] creating a high-pitched hissing sound.^[79]

Nasals (sometimes referred to as nasal stops) are consonants in which there's a closure in the oral cavity and the velum is lowered, allowing air to flow through the nose.^[80]

In an **approximant**, the articulators come close together, but not to such an extent that allows a turbulent airstream.^[79]

Laterals are consonants in which the airstream is obstructed along the center of the vocal tract, allowing the airstream to flow freely on one or both sides.^[79] Laterals have also been defined as consonants in which the tongue is contracted in such a way that the airstream is greater around the sides than over the center of the tongue.^[81] The first definition does not allow for air to flow over the tongue.

Trills are consonants in which the tongue or lips are set in motion by the airstream.^[82] The stricture is formed in such a way that the airstream causes a repeating pattern of opening and closing of the soft articulator(s).^[83] Apical trills typically consist of two or three periods of vibration.^[84]

Taps and **flaps** are single, rapid, usually **apical** gestures where the tongue is thrown against the roof of the mouth, comparable to a very rapid stop.^[82] These terms are sometimes used interchangeably, but some phoneticians make a distinction.^[85] In a tap, the tongue contacts the roof in a single motion whereas in a flap the tongue moves tangentially to the roof of the mouth, striking it in passing.

During a **glottalic airstream mechanism**, the glottis is closed, trapping a body of air. This allows for the remaining air in the vocal tract to be moved separately. An upward movement of the closed glottis will move this air out, resulting in it an **ejective consonant**. Alternatively, the glottis can lower, sucking more air into the mouth, which results in an **implosive consonant**.^[86]

Clicks are stops in which tongue movement causes air to be sucked in the mouth, this is referred to as a **velaric airstream**.^[87] During the click, the air becomes **rarefied** between two articulatory closures, producing a loud 'click' sound when the anterior closure is released. The release of the anterior closure is referred to as the click influx. The release of the posterior closure, which can be velar or uvular, is the click efflux. Clicks are used in several African language families, such as the **Khoisan** and **Bantu** languages.^[88]

Pulmonary and subglottal system [edit]

*Further information: **Breathing***

The lungs drive nearly all speech production, and their importance in phonetics is due to their creation of pressure for pulmonic sounds. The most common kinds of sound across languages are pulmonic egress, where air is exhaled from the lungs.^[89] The opposite is possible, though no language is known to have pulmonic ingressive sounds as phonemes.^[90] Many languages such as **Swedish** use them for **paralinguistic** articulations such as affirmations in a number of genetically and geographically diverse languages.^[91] Both egressive and ingressive sounds rely on holding the vocal folds in a particular posture and using the lungs to draw air across the vocal folds so that they either vibrate (voiced) or do not vibrate (voiceless).^[89] Pulmonic articulations are restricted by the volume of air able to be exhaled in a given respiratory cycle, known as the **vital capacity**.

The lungs are used to maintain two kinds of pressure simultaneously to produce and modify phonation. To produce phonation at all, the lungs must maintain a pressure of 3–5 cm H2O higher than the pressure above the glottis. However small and fast adjustments are made to the subglottal pressure to modify speech for suprasegmental features like stress. A number of thoracic muscles are used to make these adjustments. Because the lungs and thorax stretch during inhalation, the elastic forces of the lungs alone can produce pressure differentials sufficient for phonation at lung volumes above 50 percent of vital capacity.^[92] Above 50 percent of vital capacity, the **respiratory muscles** are used to "check" the elastic forces of the thorax to maintain a stable pressure differential. Below that volume, they are used to increase the subglottal pressure by actively exhaling air.

During speech, the respiratory cycle is modified to accommodate both linguistic and biological needs. Exhalation, usually about 60 percent of the respiratory cycle at rest, is increased to about 90 percent of the respiratory cycle. Because metabolic needs are relatively stable, the total volume of air moved in most cases of speech remains about the same as quiet tidal breathing.^[93] Increases in speech intensity of 18 dB (a loud conversation) has relatively little impact on the volume of air moved. Because their respiratory systems are not as developed as adults, children tend to use a larger proportion of their vital capacity compared to adults, with more deep inhales.^[94]

Source–filter theory [edit]

*Main article: **Source–filter model***

This section **needs expansion**. You can help by **adding to it**. *(February 2020)*

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The accompanying audio

The source–filter model of speech is a theory of speech production which explains the link between vocal tract posture and the acoustic consequences. Under this model, the vocal tract can be modeled as a noise source coupled onto an **acoustic filter**.^[95] The noise source in many cases is the larynx during the process of voicing, though other noise sources can be modeled in the same way. The shape of the supraglottal vocal tract acts as the filter, and different configurations of the articulators result in different acoustic patterns. These changes are predictable. The vocal tract can be modeled as a sequence of tubes, closed at one end, with varying diameters, and by using equations for **acoustic resonance** the acoustic effect of an articulatory posture can be derived.^[96] The process of inverse filtering uses this principle to analyze the source spectrum produced by the vocal folds during voicing. By taking the inverse of a predicted filter, the acoustic effect of the supraglottal vocal tract can be undone giving the acoustic spectrum produced by the vocal folds.^[97] This allows quantitative study of the various phonation types.

Perception [edit]

*Main article: **Speech perception***

Language perception is the process by which a linguistic signal is decoded and understood by a listener.^[i] To perceive speech, the continuous acoustic signal must be converted into discrete linguistic units such as **phonemes**, **morphemes**, and **words**.^[98] To correctly identify and categorize sounds, listeners prioritize certain aspects of the signal that can reliably distinguish between linguistic categories.^[99] While certain cues are prioritized over others, many aspects of the signal can contribute to perception. For example, though oral languages prioritize acoustic information, the **McGurk effect** shows that visual information is used to distinguish ambiguous information when the acoustic cues are unreliable.^[100]

While listeners can use a variety of information to segment the speech signal, the relationship between acoustic signal and category perception is not a perfect mapping. Because of **coarticulation**, noisy environments, and individual differences, there is a high degree of acoustic variability within categories.^[101] Known as the problem of **perceptual invariance**, listeners are able to reliably perceive categories despite the variability in acoustic instantiation.^[102] To do this, listeners rapidly accommodate to new speakers and will shift their boundaries between categories to match the acoustic distinctions their conversational partner is making.^[103]

Audition [edit]

*Main article: **Hearing**; for further information see **Neuronal encoding of sound***

Audition, the process of hearing sounds, is the first stage of perceiving speech. Articulators cause systematic changes in air pressure which travel as sound waves to the listener's ear. The sound waves then hit the listener's **ear drum** causing it to vibrate. The vibration of the ear drum is transmitted by the **ossicles**—three small bones of the middle ear—to the **cochlea**.^[104] The cochlea is a spiral-shaped, fluid-filled tube divided lengthwise by the **organ of Corti** which contains the **basilar membrane**. The basilar membrane increases in thickness as it travels through the cochlea causing different frequencies to resonate at different locations. This **tonotopic** design allows for the ear to analyze sound in a manner similar to a **Fourier transform**.^[105]

The differential vibration of the basilar causes the **hair cells** within the organ of Corti to move. This causes **depolarization** of the hair cells and ultimately a conversion of the acoustic signal into a neuronal signal.^[106] While the hair cells do not produce **action potentials** themselves, they release neurotransmitter at synapses with the fibers of the **auditory nerve**, which does produce action potentials. In this way, the patterns of oscillations on the basilar membrane are converted to **spatiotemporal patterns** of firings which transmit information about the sound to the **brainstem**.^[107]

Prosody [edit]

*Main article: **Prosody (linguistics)***

Besides consonants and vowels, phonetics also describes the properties of speech that are not localized to **segments** but greater units of speech, such as **syllables** and **phrases**. Prosody includes **auditory characteristics** such as **pitch**, **speech rate**, **duration**, and **loudness**. Languages use these properties to different degrees to implement **stress**, **pitch accents**, and **intonation** — for example, **stress in English** and **Spanish** is correlated with changes in pitch and duration, whereas **stress in Welsh** is more consistently correlated with pitch than duration and stress in Thai is only correlated with duration.^[108]

Theories of speech perception [edit]

Early theories of speech perception such as **motor theory** attempted to solve the problem of perceptual invariance by arguing that speech perception and production were closely linked. In its strongest form, motor theory argues that speech perception *requires* the listener to access the articulatory representation of sounds;^[109] to properly categorize a sound, a listener reverse engineers the articulation which would produce that sound and by identifying these gestures is able to retrieve the intended linguistic category.^[110] While findings such as the McGurk effect and case studies from patients with neurological injuries have provided support for motor theory, further experiments have not supported the strong form of motor theory, though there is some support for weaker forms of motor theory which claim a non-deterministic relationship between production and perception.^{[110][111][112]}

Successor theories of speech perception place the focus on acoustic cues to sound categories and can be grouped into two broad categories: abstractionist theories and episodic theories.^[113] In abstractionist theories, speech perception involves the identification of an idealized lexical object based on a signal reduced to its necessary components and normalizing the signal to counteract speaker variability. Episodic theories such as the **exemplar model** argue that speech perception involves accessing detailed memories (i.e., **episodic memories**) of previously heard tokens. The problem of perceptual invariance is explained by episodic theories as an issue of familiarity: normalization is a byproduct of exposure to more variable distributions rather than a discrete process as abstractionist theories claim.^[113]

Subdisciplines [edit]

Acoustic phonetics [edit]

*Main article: **Acoustic phonetics***

Acoustic phonetics deals with the **acoustic** properties of speech sounds. The **sensation** of sound is caused by pressure fluctuations which cause the **eardrum** to move. The ear transforms this movement into neural signals that the brain registers as sound. Acoustic waveforms are records that measure these pressure fluctuations.^[114]

Articulatory phonetics [edit]

*Main article: **Articulatory phonetics***

Articulatory phonetics deals with the ways in which speech sounds are made.

Auditory phonetics [edit]

*Main article: **Auditory phonetics***

Auditory phonetics studies how humans perceive speech sounds. Due to the anatomical features of the auditory system distorting the speech signal, humans do not experience speech sounds as perfect acoustic records. For example, the auditory impressions of **volume**, measured in decibels (dB), does not linearly match the difference in sound pressure.^[115]

The mismatch between acoustic analyses and what the listener hears is especially noticeable in speech sounds that have a lot of high-frequency energy, such as certain fricatives. To reconcile this mismatch, functional models of the auditory system have been developed.^[116]

Describing sounds [edit]

Human languages use many different sounds and to compare them linguists must be able to describe sounds in a way that is language independent. Speech sounds can be described in a number of ways. Most commonly speech sounds are referred to by the mouth movements needed to produce them. **Consonants** and **vowels** are two gross categories that phoneticians define by the movements in a speech sound. More fine-grained descriptors are parameters such as place of articulation. **Place of articulation**, **manner of articulation**, and **voicing** are used to describe consonants and are the main divisions of the **International Phonetic Alphabet** consonant chart. Vowels are described by their height, backness, and rounding. Sign language are described using a similar but distinct set of parameters to describe signs: location, movement, hand shape, palm orientation, and non-manual features. In addition to articulatory descriptions, sounds used in oral languages can be described using their acoustics. Because the acoustics are a consequence of the articulation, both methods of description are sufficient to distinguish sounds with the choice between systems dependent on the phonetic feature being investigated.

Consonants are speech sounds that are articulated with a complete or partial closure of the **vocal tract**. They are generally produced by the modification of an **airstream** exhaled from the lungs. The respiratory organs used to create and modify airflow are divided into three regions: the vocal tract (supralaryngeal), the **larynx**, and the subglottal system. The airstream can be either **egressive** (out of the vocal tract) or **ingressive** (into the vocal tract). In pulmonic sounds, the airstream is produced by the lungs in the subglottal system and passes through the larynx and vocal tract. **Glottalic** sounds use an airstream created by movements of the larynx without airflow from the lungs. **Click** consonants are articulated through the **rarefaction** of air using the tongue, followed by releasing the forward closure of the tongue.

Vowels are **syllabic** speech sounds that are pronounced without any obstruction in the vocal tract.^[117] Unlike consonants, which usually have definite places of articulation, vowels are defined in relation to a set of reference vowels called **cardinal vowels**. Three properties are needed to define vowels: tongue height, tongue backness, and lip roundedness. Vowels that are articulated with a stable quality are called **monophthongs**; a combination of two separate vowels in the same syllable is a **diphthong**.^[118] In the **IPA**, the vowels are represented on a trapezoid shape representing the human mouth: the vertical axis representing the mouth from floor to roof and the horizontal axis represents the front-back dimension.^[119]

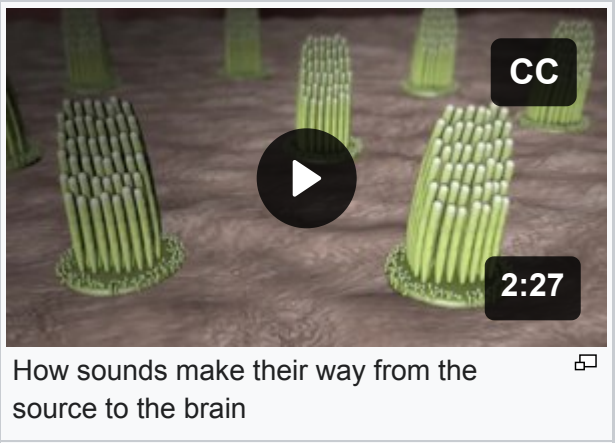
Transcription [edit]

*Main article: **Phonetic transcription***

Phonetic transcription is a system for transcribing **phones** that occur in a language, whether **oral** or **sign**. The most widely known system of phonetic transcription, the **International Phonetic Alphabet** (IPA), provides a standardized set of symbols for oral phones.^{[120][121]} The standardized nature of the IPA enables its users to transcribe accurately and consistently the phones of different languages, **dialects**, and **idiolects**.^{[120][122][123]} The IPA is a useful tool not only for the study of phonetics but also for language teaching, professional acting, and **speech pathology**.^[122]

While no sign language has a standardized writing system, linguists have developed their own notation systems that describe the handshape, location and movement. The **Hamburg Notation System** (HamNoSys) is similar to the IPA in that it allows for varying levels of detail. Some notation systems such as KOMVA and the **Stokoe system** were designed for use in dictionaries; they also make use of alphabetic letters in the local language for handshapes whereas HamNoSys represents the handshape directly. **SignWriting** aims to be an easy-to-learn writing system for sign languages, although it has not been officially adopted by any deaf community yet.^[124]

Sign languages [edit]



Unlike spoken languages, words in **sign languages** are perceived with the eyes instead of the ears. Signs are articulated with the hands, upper body and head. The main articulators are the hands and arms. Relative parts of the arm are described with the terms **proximal** and **distal**. Proximal refers to a part closer to the torso whereas a distal part is further away from it. For example, a wrist movement is distal compared to an elbow movement. Due to requiring less energy, distal movements are generally easier to produce. Various factors – such as muscle flexibility and being considered **taboo** – restrict what can be considered a sign.^[125] Native signers do not look at their conversation partner's hands. Instead, their gaze is fixated on the face. Because **peripheral vision** is not as focused as the center of the visual field, signs articulated near the face allow for more subtle differences in finger movement and location to be perceived.^[126]

Unlike spoken languages, sign languages have two identical articulators: the hands. Signers may use whichever hand they prefer with no disruption in communication. Due to universal neurological limitations, two-handed signs generally have the same kind of articulation in both hands; this is referred to as the Symmetry Condition.^[125] The second universal constraint is the Dominance Condition, which holds that when two handshapes are involved, one hand will remain stationary and have a more limited set of handshapes compared to the dominant, moving hand.^[127] Additionally, it is common for one hand in a two-handed sign to be dropped during informal conversations, a process referred to as weak drop.^[125] Just like words in spoken languages, coarticulation may cause signs to influence each other's form. Examples include the handshapes of neighboring signs becoming more similar to each other (**assimilation**) or weak drop (an instance of **deletion**).^[128]

See also [edit]

- Motor theory of speech perception**
- Exemplar theory**
- Articulatory phonology**

References [edit]

Notes [edit]

- ↑ Linguists debate whether these stages can interact or whether they occur serially (compare **Dell & Reich (1981)** and **Motley, Camden & Baars (1982)**). For ease of description, the language production process is described as a series of independent stages, though recent evidence shows this is inaccurate.^[12] For further descriptions of interactive activation models see **Jaeger, Furth & Hilliard (2012)**.
- ↑ or after part of an utterance has been planned; see **Gleitman et al. (2007)** for evidence of production before a message has been completely planned
- ↑ adapted from **Sedivy (2019**, p. 411) and **Boersma (1998**, p. 11)
- ↑ See **Feldman (1966)** for the original proposal.
- ↑ See **#The larynx** for further information on the anatomy of phonation.
- ↑ Hawaiian, for example, does not contrast voiced and voiceless plosives.
- ↑ There are languages, like **Japanese**, where vowels are produced as voiceless in certain contexts.
- ↑ See **#Articulatory models** for further information on acoustic modeling.
- ↑ As with speech production, the nature of the linguistic signal varies depending on the **language modality**. The signal can be acoustic for oral speech, visual for signed languages, or tactile for manual-tactile sign languages. For simplicity acoustic speech is described here; for sign language perception specifically, see **Sign language#Sign perception**.

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